UNIVERSITY OF BOTSWANA

2011/2012 – EXAMINATIONS

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COURSE NUMBER MAT111 **DURATION** 2 hrs **DATE** NOV 2011
TITLE OF PAPER INTRODUCTORY MATHEMATICS I
SUBJECT MATHEMATICS **TITLE OF EXAMINATION** BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER **ALL** QUESTIONS IN SECTION **A** AND ANY **TWO(2)** QUESTIONS FROM SECTION **B**.
- ALL MARKS ARE INDICATED IN BRACKETS [ ].

NUMBER OF PAGES INCLUDING COVER PAGE: $\boxed{5}$

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Section A

Answer All Questions: Each Question carries 10 Marks

Question 1

(a) Write $0.2\overline{17}$ in the form $\frac{p}{q}$ where p and q are integers. [3]

(b) Simplify $\quad \dfrac{x^7x^{-4}}{x^{-5}x^3}$ [3]

(c) Solve $\quad 3^{x+1} = 4^{x-1}$ [4]

Question 2

Solve the following:

(a) $x(x+1) \leq -2(2x+3)$ [6]

(b) $|2x-3| = |4x-4|$ [4]

Question 3

(a) Given that $f(x) = x^2 - 1$, and $g(x) = \dfrac{1}{x+1}$,
find $(fg)(x)$ and $(\dfrac{f}{g})(x)$ and their domains. [6]

(b) Find the equation of the line that passes through (2,5) and is perpendicular to $3x - 2y = 0$. [4]

Question 4

a) Solve for x, $\quad \dfrac{1}{3}\log(x^6) = 5\log 3 - \log 27$ [4]

b) Solve the equation $\quad x^3 - 4x^2 + x + 6 = 0$ [6]

Question 5

(a) Find the value of $\sin\dfrac{2\pi}{3}$ without using calculator [4]

(b) Prove the identity $\cos 2x = \cos^4 x - \sin^4 x$. [4]

(c) Evaluate $\sin^{-1} 1$ without using calculator [2]

Question 6

(a) Express the complex number

$$\frac{1}{1+i}$$

in polar form. [5]

(b) Solve $x^2 - x + 1 = 0$. [5]

Section B

Answer TWO(2) Questions OUT of FOUR:
Each Question carries 20 Marks

Question 7

(a) Solve $\sqrt{x} - \sqrt{x-2} = 1$ [6]

(b) Solve for x, $3(3^x) - 2(3^{-x}) + 5 = 0$ [8]

(c) If α and β are roots of the equation $x^2 + 3x + 1$, find the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ [6]

Question 8

(a) Solve $4 < |2x-1| \leq 6$ [6]

(b) Given that $f(x) = 1 - x^2$, $x \in I\!R$ and $g(x) = \dfrac{2}{3-4x}$, $x \in I\!R$, $x \neq \dfrac{3}{4}$,

(i) Find a formula for $g \circ f$ and the domain of $g \circ f$. [5]

(ii) Find a formula for $g^{-1}(x)$ and its domain. [5]

(c) Define what is meant by an odd function and an even function.
Show that if f and g are odd functions then $h(x) = f(x)g(x)$ is an even function. [4]

Question 9

a) Sketch the graph of $f(x) = x^2 + 4x$
and also sketch the graph of $g(x) = (x-1)^2 + 4(x-1) + 3$ on the same axis. [8]

b) Given that $12\log(\frac{1}{y}) = 4\log(\frac{1}{x}) - 3\log(y^2)$, find y in terms of x. [5]

c) Find the remainder when $f(x) = x^3 - 2x^2 - 5x + 6$ is divided by $(x+2)$ hence factorize $f(x)$. [7]

Question 10

(a) Express $\sin 5x - \sin 3x$ as a product.
Hence or otherwise solve the equation $\sin 5x - \sin 3x = 0$. [8]

(b) Solve the equation $z^3 = i$, giving solutions in polar form. [8]

(c) Express $\cos 3\theta$ in terms of $\sin\theta$ and $\cos\theta$ [4]